

Framing, not structure: an invariance audit of large language model cooperation in the iterated prisoner’s dilemma

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Abstract

Large language models increasingly act as agents that negotiate and coordinate, making their cooperation an AI governance question. We ask not how much they cooperate but whether what we call cooperation is a property of the game at all. Using a fully crossed corpus of 2,400 iterated prisoner’s dilemma dyads across four frontier models, three payoff magnitudes, five languages and four persona pairings, we test two manipulations whose irrelevance is a theorem rather than a hypothesis. Multiplying every payoff by a positive constant leaves the game strategically identical, yet moves cooperation by up to 0.24; changing the prompt language with objective framing held fixed moves it by up to 0.15. Stranger still: agents told they are cooperative open cooperatively 31% of the time, while agents told they are selfish do so 91%, in every model and language. This inversion, the direction of the payoff-scale effect and the shape of the memory-one fingerprints are jointly explained if the models read the stated penalties as rewards, pattern-matching the matrix to a canonical prisoner’s dilemma rather than solving the stated problem. Cooperation rates and strategy labels reported without an invariance audit may describe the prompt rather than the agent.

Keywords: machine behaviour; large language models; iterated prisoner’s dilemma; cooperation; framing effects; evolutionary game theory; multi-agent systems

1 Introduction

Large language models (LLMs) are increasingly deployed as *agents* in recommendation systems, negotiation tools and multi-agent assistants. In these settings they face cooperation dilemmas in which behaviour emerges from strategic interaction rather than from isolated model outputs, and the resulting behaviour is shaped by training, prompting, role assignment and linguistic framing, with direct implications for safety and governance [1, 2]. The proposal that machines be studied as behaving entities, with the observational tools of ethology and the experimental discipline of the behavioural sciences, has become considerably more urgent as the population of interacting artificial agents has grown [1].

A rapidly expanding literature evaluates LLMs through game-theoretic lenses, particularly via matrix and repeated games, revealing systematic departures from Nash equilibria, persistent cooperative biases and sensitivity to contextual factors such as language and incentives [3–7]. Models have been probed with the instruments of cognitive psychology [8] and used as

simulated economic subjects [9–11]. The prisoner’s dilemma is the natural testbed: a two-by-two matrix with $T > R > P > S$ and $2R > T + S$ generates a well-mapped strategic landscape in which reciprocity, forgiveness and exploitation are defined precisely and their evolutionary fates computed [12–16], a landscape that underpins a large body of work on the statistical physics of cooperation [17–19] and on mechanisms such as intention recognition and commitment that shift the cooperative equilibrium [20, 21].

Most existing evaluations report one of two things: a scalar, that model X cooperates at rate p ; or a label, that model X plays tit-for-tat. Both presuppose something that has not been established. A cooperation rate is a property of an agent only if it is stable under transformations that do not change the problem. A strategy label is informative only if the behaviour it names is actually generated by something in the family from which the label is drawn. Neither presupposition is safe for a system whose input is a natural-language string and whose behaviour is known to depend on the surface form of that string in ways carrying no task content [22–25].

This paper takes those presuppositions as the object of study. We adopt two invariance principles as the test. The first is the affine invariance of von Neumann–Morgenstern utility: multiplying every payoff by the same positive constant leaves best replies, equilibria and evolutionary dynamics unchanged [26, 27]. The second is the invariance of a game to the language in which its rules are stated. These give manipulations with an unusual property for behavioural work — the null hypothesis is a theorem, not a convenient default — so that any movement is a failure rather than merely an observation.

The present study is a revision and extension of our earlier analysis of the same corpus [28], which established that LLM cooperation rises with payoff magnitude in direct contradiction to an evolutionary game theory (EGT) baseline, and attributed the divergence to alignment training and the cooperative bias of human training data. Here we retain that empirical result, add a fourth model, and ask a prior question: whether the movement is a strategic response at all. Three questions organise the work:

1. Do LLM agents satisfy the invariances that any game-theoretic description of them would require, and does the answer differ across models and languages?
2. How much of the variance in cooperation does the experimental design explain, relative to the variance between replicates of an identical prompt?
3. Is the resulting play generated by a strategy in the standard sense, and what mechanism accounts jointly for the invariance failures?

The contributions are: **(i)** an invariance audit of payoff magnitude and prompt language in which the null is exact, including a language test that holds the objective framing fixed and thereby separates a language effect from a framing effect (§3.2–§3.3); **(ii)** the finding that the assigned persona inverts, moving behaviour opposite to its semantic content at the opening move in every model and every language (§3.4); **(iii)** a single account — that the models read the stated penalties as rewards — which explains the persona inversion, the direction of the payoff-scale effect and the shape of the memory-one fingerprints together (§3.5); and **(iv)** a deductive audit of strategy attribution showing that in half to seven-eighths of agent-games no memory-one rule fits at all, so that an archetype label reports the resolution of the classifier rather than the behaviour of the model (§3.6).

2 Material and methods

2.1 Framework

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [29] provides the computational infrastructure. Experimental conditions are defined via JSON configuration files specifying payoff structures, game horizon, LLM backends and languages; at runtime FAIRGAME combines these with language-specific prompt templates, simulates repeated normal-form games and logs round-by-round trajectories. We extend it with payoff scaling for the prisoner’s dilemma.

2.2 Stage game and payoff framing

Agents played a symmetric two-player prisoner’s dilemma (figure 1a). The templates state the cell values as *penalties* — years of imprisonment in the prisoner’s-dilemma narrative — and instruct the agent to *minimise* them, so a smaller number is a better outcome. Under that framing the base cells are $T = 0$, $R = 2$, $P = 6$, $S = 10$, satisfying $T < R < P < S$, which is the penalty-space image of the usual $T > R > P > S$. Option A is therefore the dominant, defecting action and Option B is cooperation: mutual cooperation carries the lowest symmetric penalty (2, 2), mutual defection (6, 6), and defecting on a cooperator carries none at all. On the utility scale $u = -\text{penalty}$ the two normalised descriptors of the dilemma are greed $(T - R)/(T - S) = 0.20$ and fear $(P - S)/(T - S) = 0.40$.

We flag this polarity explicitly because it is easy to invert and because inverting it maps every cooperation rate $x \mapsto 1 - x$ and reverses the sign of both the payoff-scale and the endgame effects. Our analysis pipeline reproduces the cooperation rates of Huynh et al. [28] to a mean absolute error of 0.002 under the mapping above, against 0.171 under the inverted one.

Every cell was multiplied by a payoff scale $\lambda \in \{0.1, 1, 10\}$. Because $\lambda > 0$ this is a positive affine rescaling: the outcome ordering, the best-reply correspondence and the equilibrium set are unchanged, so the three games are the same game presented with different numbers. The realised scale was recovered per game from the payoff values actually recorded rather than assumed from the directory: 80 rounds (0.17% of the corpus) carry $\times 1$ payoffs inside a folder labelled $\times 10$, and are labelled by their realised scale throughout.

2.3 Design and prompting

The design crossed four models — Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large — with three payoff scales, five prompt languages (English, French, Vietnamese, Chinese, Arabic), four persona pairings and ten replicates, giving 600 dyads per model and 2,400 in total (figure 1b). Each dyad ran for ten rounds and both agents are recorded, so the corpus contains 4,800 agent-games and 48,000 binary decisions, with exactly 40 dyads in every model $\times$ scale $\times$ language cell.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players’ choices, and returned a single token naming one of two options. Communication was disabled, so the only channel between agents is the action history. Actions were labelled **OptionA** and **OptionB** in all languages, avoiding the connotations that the words “cooperate” and “defect” would carry. Each agent received a system-prompt persona, either *cooperative* or *selfish*; an agent was never told its opponent’s persona, so the opponent’s persona is a manipulation the focal agent can learn about only through play.

Decoding used a fixed low token budget with reasoning traces disabled. This is not cosmetic: models that emit hidden reasoning can exhaust a small output budget inside the trace and return a truncated string that fails to parse, and a harness that then falls back to a default action produces an apparent cooperation rate of 1.0 that is an artefact of the parser. Parse-failure and fallback rates were logged for every cell and required to be below 0.02.

2.4 Behavioural measures

The primary measure is the cooperation rate, the proportion of a game’s ten rounds on which an agent chose Option B. Because both agents of every dyad appear as rows, the exploited (CD) and exploiting (DC) shares are equal by construction and are merged into a single anti-coordination band in composition analyses.

The memory-one fingerprint is the vector $(p(C | R), p(C | S), p(C | T), p(C | P))$ of probabilities of cooperating after each joint outcome of the previous round. This is the complete description of a memory-one strategy and places every canonical rule at a known point: AllC at $(1, 1, 1, 1)$, AllD at $(0, 0, 0, 0)$, tit-for-tat at $(1, 0, 1, 0)$, win-stay lose-shift at $(1, 0, 0, 1)$ and grim trigger at $(1, 0, 0, 0)$. Niceness is $p(C)$ on round one and reciprocity is $p(C | C_{-1}) - p(C | D_{-1})$.

2.5 Strategy read-out

Assigning a strategy label was done in two stages, kept distinct because they license different claims. The first is exact and deductive: for each agent-game we enumerated all 32 deterministic memory-one rules and retained those consistent with every observed transition. A game may be matched by exactly one rule, by several, or by none, and only the first case supports the statement that the agent played that rule. The second is inductive and applies only where the first fails: a long short-term memory classifier [30] trained on synthetic play from known generating strategies returns a posterior over archetypes, whose argmax we take as the nearest neighbour. We also record, for each agent-game, the minimum number of rounds on which play departs from its own nearest rule.

2.6 Statistical procedures

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [31]. This is consequential: rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would understate the interval by roughly $\sqrt{2}$. The payoff-scale slope is the change in cooperation per decade of λ , fitted on $\log_{10} \lambda$. Language disparity is the difference between a model’s highest- and lowest-cooperation language, tested against a permutation null that shuffles the language label across dyads within a model over 5,000 draws. The variance decomposition is sequential (Type I) in the order model, payoff scale, language, persona; entering model identity first makes its share an upper bound.

Three reliability checks were run whose expected answer is “nothing here”. Agents 1 and 2 of a matched pairing receive identical prompts and their cooperation rates correlate at $r = 0.67$ – 0.95 with a mean position bias of $+0.017$. The ten replicates of a cell share a prompt and their spread is 1.3–1.6 times what independent rounds would produce. Cell means reproduce across a random split of the dyads with Spearman–Brown reliability of 0.90–0.97.

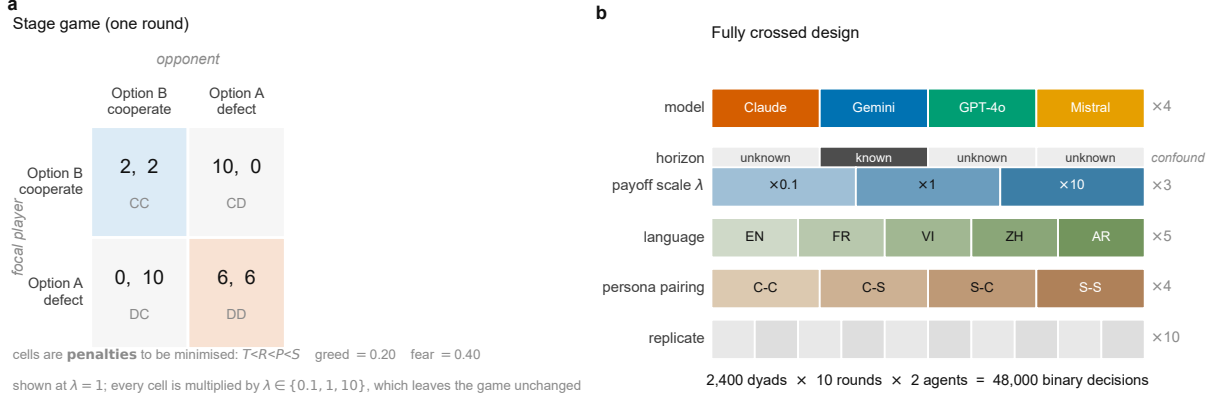


Figure 1: The game and the design. (a) The stage game at $\lambda = 1$. Cells are penalties the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Greed and fear are computed on the utility scale $u = -\text{penalty}$. Every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$, a positive affine rescaling that leaves the game strategically identical. (b) The fully crossed design: 2,400 dyads, 48,000 binary decisions, with 40 dyads in every model $\times$ scale $\times$ language cell. The horizon row is shaded as a confound rather than a factor, because only the Gemini arm was told the round count.

2.7 Declared confound

The Gemini arm was run with the round count disclosed to the agents; the Claude, GPT-4o and Mistral arms with the horizon hidden. The Gemini column is therefore “model $\times$ horizon” and is marked as such in figure 1b. A second issue, the fact that the templates do not state the objective identically in every language, is a result rather than a nuisance and is treated in §3.3.

3 Results

3.1 Aggregate cooperation conceals more than it reports

Pooled across the design, the four models cooperate at very different rates: 0.74 for Gemini 3.5 Flash-Lite, 0.63 for Claude 3.5 Haiku, 0.52 for Mistral Large and 0.45 for GPT-4o. Read naively this is the headline of a conventional benchmark report, and it is the form in which such results are usually communicated.

It is worth establishing immediately how little that number carries. The distribution of within-game cooperation is not concentrated near its mean for any model. Mistral Large plays 25.0% of its games entirely cooperatively and 24.3% entirely defecting, so its mean of 0.52 describes a bimodal mixture rather than a typical game; Gemini reaches 36.9% wholly cooperative games and GPT-4o 14.9% wholly defecting ones.

How much of this spread does the design account for? Very little. In a sequential decomposition, model identity explains 9.5% of the variance in dyad-level cooperation, persona pairing 5.5%, payoff scale 3.6% and language 0.2%, for a total of 18.7%. The remaining 81.3% is within-cell: ten replicates of a byte-identical prompt disagree with one another more than all four deliberate manipulations together move the mean. Because model identity was entered first, its 9.5% is an upper bound. This is the context in which the invariance tests should be read: we are not asking whether the models differ, but whether the differences that manipulations produce are differences a strategic agent could have. Table 1 summarises the per-model profile developed below.

Table 1: Behavioural profile of each model. Cooperation is pooled over the whole design. The payoff-scale slope is the change in cooperation per decade of λ with 95% bootstrap CI; its null is exactly zero. The language gap is measured within the three languages sharing one objective framing. “Opening C” gives $p(\text{cooperate on round 1})$ for an agent told it is cooperative and for one told it is selfish; the ordering is reversed in every model. Distance is Euclidean, in the four-dimensional memory-one fingerprint space, to the nearest canonical rule, which are themselves at most 2.0 apart. “Exact rule” is the share of agent-games in which exactly one of the 32 deterministic memory-one rules reproduces the whole trajectory.

Model	Coop.	Scale slope	Lang. gap	Opening C		Nearest	Exact
	rate	(per decade)	(fixed frame)	coop.	selfish	rule (dist.)	rule
Claude 3.5 Haiku	0.63	+0.120 [0.102, 0.137]	0.087**	0.14	0.81	TFT (0.76)	9.7%
Gemini 3.5 Flash-Lite	0.74	+0.030 [0.007, 0.053]	0.052	0.54	0.98	GRIM (0.90)	21.9%
GPT-4o	0.45	+0.117 [0.094, 0.140]	0.028	0.38	0.84	GRIM (0.71)	21.8%
Mistral Large	0.52	+0.015 [−0.012, 0.043]	0.152***	0.19	1.00	GRIM (1.01)	48.7%

** $P < 0.01$, *** $P < 0.001$ against a permutation null; unmarked gaps do not exceed their null.

3.2 Payoff magnitude changes behaviour

Multiplying every payoff by a positive constant leaves best replies, equilibria and evolutionary dynamics unchanged, so the null for a strategic agent is not approximately zero but exactly zero.

Two of the four models violate it decisively (figure 2a). Claude 3.5 Haiku gains 0.120 cooperation per decade of λ (95% CI [0.102, 0.137]) and GPT-4o gains 0.117 ([0.094, 0.140]), total swings of 0.24 and 0.23. Gemini shows a smaller but non-zero slope of 0.030 ([0.007, 0.053]), and only Mistral Large is consistent with invariance at 0.015 ([−0.012, 0.043]).

The direction reproduces the central empirical result of Huynh et al. [28] on this corpus: cooperation *rises* with the stated stakes, where an evolutionary account of the same matrix predicts that defection should take over as stakes grow. What the invariance framing adds is that the movement cannot be a strategic response in either direction, because there is no strategic difference to respond to. Note also that the effect is not monotone in the way a magnitude heuristic would predict: for Claude and GPT-4o almost the entire swing occurs between $\times 0.1$ and $\times 1$, with $\times 1$ and $\times 10$ nearly indistinguishable, which points to sensitivity to the appearance of the numbers — fractional versus integral, single- versus multi-digit — rather than to magnitude as such.

3.3 Prompt language changes behaviour, with framing held fixed

Measured across all five languages, three of four models show a language disparity beyond its permutation null: 0.156 for Gemini ($P < 0.001$), 0.155 for Mistral ($P < 0.001$) and 0.087 for Claude ($P = 0.001$), against 0.040 for GPT-4o ($P = 0.70$).

That comparison is confounded, and we treat the confound as a result. The English, French and Vietnamese templates instruct the agent to minimise a penalty, whereas the Arabic and Chinese templates instruct it to maximise a reward while still calling the outcomes penalties. A raw five-language comparison therefore mixes a language manipulation with an objective-framing manipulation, and the language effect must be re-tested with framing fixed.

Doing so within English, French and Vietnamese alone (figure 2b, solid bars) leaves the effect intact for two of four models: Mistral Large retains a disparity of 0.152 ($P < 0.001$) and

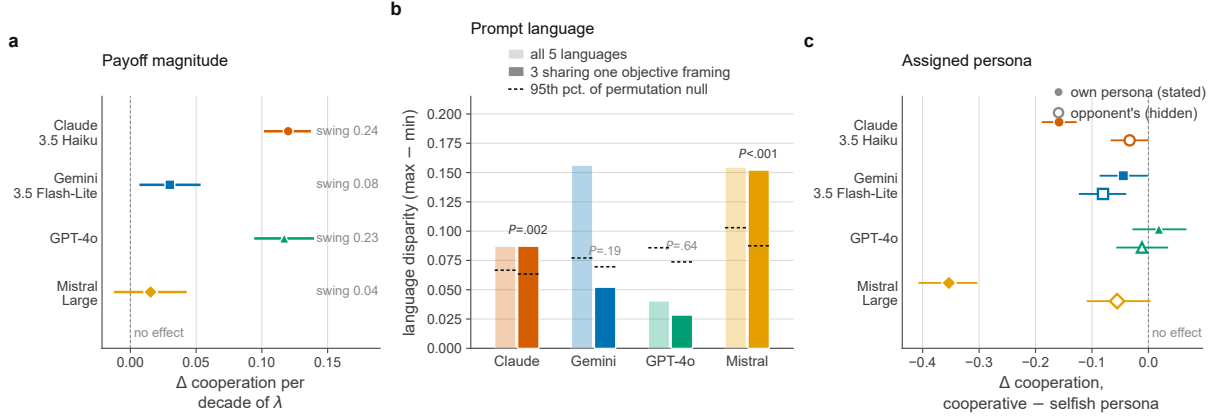


Figure 2: Three manipulations that cannot matter, and all three do. (a) Change in cooperation per decade of the payoff scale λ , with 95% bootstrap confidence intervals resampling whole dyads. A positive affine rescaling leaves the game strategically identical, so the null is exactly zero. Claude and GPT-4o depart from it by more than 0.11 per decade; the annotation gives the total swing across the three scales. (b) Language disparity, the gap between a model’s highest- and lowest-cooperation language. Pale bars use all five languages, in which objective framing varies with language; solid bars use only English, French and Vietnamese, which share a single framing. Dashes mark the 95th percentile of a permutation null; P values are for the framing-controlled comparison. (c) Effect of the assigned persona. Filled markers give the difference between an agent told it is cooperative and one told it is selfish; open markers give the same difference for the *opponent’s* persona, which is never disclosed. Three of four filled markers lie left of zero: the instruction moves behaviour opposite to its semantic content.

Claude 0.087 ($P = 0.002$), both above their permutation nulls. Gemini’s apparent disparity collapses from 0.156 to 0.052 and ceases to be significant ($P = 0.19$), identifying most of its five-language gap as a framing effect rather than a language effect. GPT-4o remains invariant on both readings ($P = 0.64$). The framing contrast itself is small and inconsistent in sign across models (-0.093 to $+0.064$), so it does not act as a simple main effect either.

The surviving effects are substantial: Mistral cooperates at 0.61 in English and 0.46 in Vietnamese — three languages, one objective framing, one payoff matrix, a gap of 0.15. We emphasise that this is a claim about two of four models rather than three, and that the more cautious claim is the one the data support. Because the disparity is a range, it is invariant under $x \mapsto 1 - x$ and so is the one result in this paper that does not depend on the polarity convention of §2.2 at all.

3.4 The assigned persona inverts

The persona manipulation differs from the previous two in that it is not irrelevant by theorem; a persona could in principle be read as information. But an agent’s own persona carries no information about the game, and the four pairings present identical payoff matrices. What we find is not that the persona fails to move behaviour, but that it moves it the wrong way.

Pooled over the corpus, an agent told it is cooperative cooperates 0.517 of the time and an agent told it is selfish 0.651. The effect is concentrated in the opening move and is then eroded by the interaction: at round one the two groups cooperate at 0.312 and 0.908 respectively, a gap of -0.60 ; by round ten the gap has closed to $+0.01$. The inversion at the opening holds in every model — Claude 0.14 against 0.81, Mistral 0.19 against 1.00, GPT-4o 0.38 against 0.84, Gemini

0.54 against 0.98 — in every language, with gaps from -0.42 to -0.73 , and at every payoff scale, with gaps of -0.81 , -0.49 and -0.48 at $\lambda = 0.1, 1, 10$. Mistral Large told it is selfish opens cooperatively in 100% of its 600 games; told it is cooperative, in 19%.

At the level of the whole game the pooled effect remains negative for three of four models (figure 2c): -0.354 for Mistral ($[-0.408, -0.303]$, $d = -0.94$), -0.158 for Claude ($[-0.189, -0.127]$, $d = -0.61$) and -0.045 for Gemini ($[-0.086, -0.001]$), with GPT-4o indistinguishable from zero ($+0.018$, $[-0.028, 0.067]$).

One further observation complicates any reading in which persona simply installs a disposition. The opponent’s persona — never disclosed, inferable only from play — has a detectable effect for Gemini, which cooperates 0.080 less against a cooperative opponent, and marginally for Claude. For Gemini this indirect effect is larger than its own-persona effect, which is the signature of an agent responding to the interaction rather than to the instruction.

3.5 One account of all three: the frame is misread

The three failures above look like three separate sensitivities. They are jointly explained by a single hypothesis: that the models treat the displayed numbers as rewards to be maximised rather than as penalties to be minimised, pattern-matching the matrix to the canonical reward-framed prisoner’s dilemma in which the largest entry is the prize.

Under a reward reading of the same table, Option A yields the symmetric pair (6, 6) and Option B can yield the single largest number, 10. An agent told it is cooperative and reaching for the outcome that looks mutually good would therefore choose A; an agent told it is selfish and reaching for the largest attainable number would choose B. Both are exactly what we observe, and both are the opposite of what the stated objective implies, because under minimisation A is the dominant defecting action and the 10 is the sucker’s penalty. The hypothesis also predicts the sign of the payoff-scale effect: enlarging λ magnifies the numbers and makes the salient maximum more conspicuous, so the share of B should rise with λ , which is what figure 2a shows.

The memory-one fingerprints carry the same signature (figure 3b). All four models cooperate readily after having been exploited — $p(C | S)$ is 0.48, 0.81, 0.63 and 0.96 for Claude, Gemini, GPT-4o and Mistral — where tit-for-tat, grim and win-stay lose-shift all prescribe 0. Being “exploited” in this corpus means receiving the penalty 10, the largest number on the table; an agent that has just been handed the biggest number and repeats the action that produced it is behaving consistently under a reward reading and inexplicably under the stated one. Symmetrically, three of four cooperate *less* after having successfully exploited the opponent, where the payoff was 0: $p(C | T)$ is 0.34, 0.14 and 0.11 for Gemini, GPT-4o and Mistral against tit-for-tat’s 1. Mistral’s reciprocity index is accordingly negative, -0.149 .

We present this as the most parsimonious account rather than as a demonstration. A competing explanation is ordinal: “Option A” is listed first, and a cooperative persona may be reaching for the default rather than for the larger symmetric number. Distinguishing the two requires a manipulation we did not run — re-labelling or reordering the options, or restating the identical matrix in explicit reward form and testing whether behaviour mirrors — and we regard that as the natural next experiment. What the present data do establish is that the stated objective is not what the models are optimising.

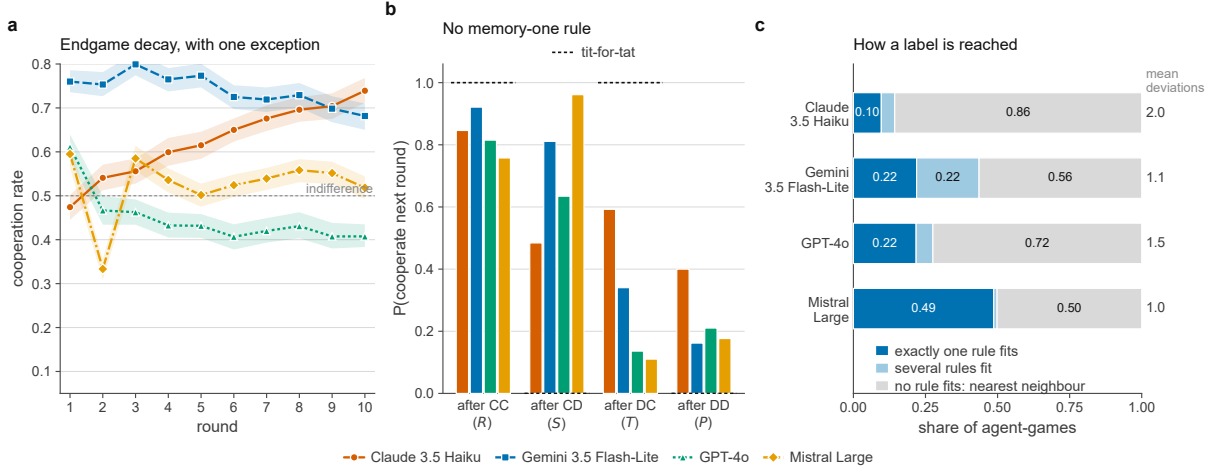


Figure 3: What the play is not. (a) Cooperation by round with 95% bootstrap bands. Three of four models decay towards the endgame, consistent with backward induction; Claude is the exception, rising from 0.47 to 0.74. Gemini is the single arm that was told the round count. (b) The memory-one fingerprint: probability of cooperating after each joint outcome of the previous round, the complete description of a memory-one strategy. Dashes mark tit-for-tat. All four cooperate readily after being exploited (CD), where every canonical rule except AllC prescribes 0, and three of four cooperate less after exploiting (DC) than after being exploited. (c) How an archetype label is actually reached. Dark: exactly one of the 32 deterministic memory-one rules reproduces the whole trajectory. Mid: several do. Grey: none does, and the label is a nearest-neighbour statement. The right-hand column gives the mean number of rounds out of ten on which play departs from its own nearest rule.

3.6 The play is not a strategy

Endgame. With a finite horizon, backward induction predicts decay towards mutual defection. Three of four models show it (figure 3a): GPT-4o falls from 0.61 to 0.41 between rounds one and ten (-0.20), Gemini from 0.76 to 0.68 (-0.08) and Mistral from 0.60 to 0.52 (-0.08). Claude is the exception, rising from 0.47 to 0.74 ($+0.27$). This reproduces the endgame pattern reported for the same corpus by Huynh et al. [28], including the Claude exception.

No memory-one rule. The fingerprints place every model far from every canonical strategy. The nearest archetype is tit-for-tat for Claude at Euclidean distance 0.76, and grim trigger for GPT-4o (0.71), Gemini (0.90) and Mistral (1.01). For scale, the canonical rules are at most 2.0 apart from one another in this space, so a distance of 0.7–1.0 is not a near miss.

A label is mostly not a description. Exactly one of the 32 deterministic memory-one rules reproduces the entire trajectory in 9.7% of Claude’s agent-games, 21.8% of GPT-4o’s, 21.9% of Gemini’s and 48.7% of Mistral’s (figure 3c). In the remaining 86%, 72%, 56% and 50% respectively no rule in the class fits and the label is a nearest-neighbour statement; where a label is assigned this way, play departs from the rule it names on 1.0 to 2.0 rounds out of ten. A further caveat limits even the deductive fraction: ten rounds rarely visit all four conditioning states, so rules differing only in an unvisited state cannot be distinguished, and coverage figures for the wider class are statements about a *set* of rules rather than about a rule.

3.7 What is stable

It would be a misreading to conclude that nothing here is measurable. The quantities we compare are reliable even though individual games are not: split-half reliability of the cell means is 0.90 for Claude, 0.91 for GPT-4o, 0.95 for Gemini and 0.97 for Mistral. What is unstable is the individual game. Ten replicates of an identical prompt have a standard deviation 1.3 to 1.6 times what independent rounds would produce. The correct summary is that these systems have reproducible *propensities* and irreproducible *episodes*.

4 Discussion

4.1 What an invariance failure means

The central results are failures of two invariances that are not empirical regularities but consequences of how games are defined. Two of four models move by more than 0.11 in cooperation per decade of payoff scale, and two of four move by 0.09–0.15 across languages sharing an objective framing. The consequence is not that the models are irrational in the ordinary sense of choosing dominated actions; it is more specific. The object that a cooperation rate is supposed to measure is not well defined for these systems, because the measurement changes when the thing measured does not. Reporting that a model cooperates at 0.45 without reporting that the same model cooperates at 0.29 or 0.53 depending on whether the payoffs are written as decimals or as hundreds is not a partial description but a description of an artefact of presentation.

This connects the machine-behaviour programme [1] directly to the framing literature in human decision making [32, 33]. The disanalogy is instructive. Human framing effects are systematic, map to a compact set of psychological mechanisms and reproduce across populations. What we observe is heterogeneous across models — Mistral is scale-invariant but the most persona-inverted; GPT-4o is language-invariant and persona-inert but strongly scale-sensitive — and sits on top of a within-cell variance that dwarfs it. There is no single “LLM framing effect” to characterise, only a per-model, per-manipulation profile that must be measured.

4.2 Reinterpreting the cooperative bias

Our earlier analysis of this corpus [28] read the rise of cooperation with stakes as evidence of alignment training and of the cooperative bias of human training data, the LLM distribution pulling away from an EGT baseline in the prosocial direction. The persona inversion suggests a complementary and less flattering mechanism. If the models are reading the penalties as rewards, then part of what has been measured as prosociality is number-chasing: the action that registers as “cooperation” under the correct framing is also the action carrying the largest single entry on the table, and it becomes more attractive precisely as λ makes that entry larger.

We do not claim this displaces the alignment account. The two are separable by experiment — restate the identical game in explicit reward form, in which the cooperative action no longer carries the largest number, and see whether the stake effect survives — and we regard that as the most valuable single extension of this work. What the present data do establish is that a cooperation rate measured under a penalty framing cannot be read as prosociality without that control.

4.3 Consequences for strategy attribution and for hybrid populations

A large part of the appeal of the prisoner’s dilemma is that its strategy space is well mapped, and it is tempting to import that map wholesale. Our results suggest the import is not licensed. A statement of the form “model X plays tit-for-tat” should be accompanied by the fraction of agent-games in which a rule was deductively established rather than assigned by nearest neighbour, and by the number of rounds on which play departs from its assigned label. On our corpus those numbers are 10–49% and 1.0–2.0 rounds out of ten.

The results also bear on efforts to model populations in which artificial and human agents interact [2, 34, 35]. Such models are typically built by assigning agents strategies from a known family and computing the resulting dynamics. If an artificial agent’s cooperation depends on the units in which payoffs are written, its behaviour is not a function of the game and cannot be carried across a replicator equation that only knows the game: two population models identical up to a rescaling would make different predictions about the same agent. And if 81% of the variance is within-cell, the appropriate representation is not a strategy but a distribution over strategies, possibly one whose support changes with framing.

4.4 Limitations

Four limitations bound these conclusions. The model set is the most consequential: four models is a small sample of a fast-moving population, and nothing here establishes that the pattern generalises to models with explicit reasoning procedures, which may well perform the minimisation the present models appear to skip. Testing the invariances on reasoning models is the single most valuable extension, and we regard these results as establishing the diagnostic rather than settling the question.

The frame-misreading account is an inference from converging evidence, not a controlled demonstration; the option-ordering alternative in §3.5 remains open until the relabelling experiment is run.

The horizon confound remains: the Gemini arm knew the round count and the others did not, so that column mixes model identity with horizon knowledge. And the framing-controlled language analysis rests on three languages rather than five, which reduces power, so the two null results should be read with that in mind. Finally, ten rounds is short: it under-identifies the memory-one vocabulary and compresses any endgame effect into a small number of rounds.

4.5 Conclusion

Four frontier language models playing 2,400 iterated prisoner’s dilemmas fail invariances that any game-theoretic description of them would require. Rescaling payoffs, which cannot change the game, moves cooperation by up to 0.24. Changing the prompt language with the objective framing held fixed moves it by up to 0.15. And the assigned persona moves behaviour opposite to its semantic content: told they are cooperative, agents open cooperatively 31% of the time; told they are selfish, 91%. A single account — that the models read the stated penalties as rewards — covers the inversion, the direction of the scale effect and the shape of the memory-one fingerprints.

We therefore propose a minimal standard. A cooperation rate reported for a language model should be accompanied by its sensitivity to payoff scale and to prompt language; a strategy label

should be accompanied by the fraction of games in which that label was deductively established; and any study framing its payoffs as penalties should verify that the models are optimising the stated objective before reading the result as prosociality. These are cheap to compute and they change what the numbers mean. Until they are routine, the field risks building an evolutionary game theory of artificial agents on quantities that are properties of prompts rather than of agents.

Data accessibility

The full corpus of 2,400 dyads, the analysis pipeline that produces every figure and table, and the supplementary figure suite are available at [REPOSITORY URL]. Every number appearing in a figure is also written to a machine-readable CSV file in the same repository. A permanent archive with a DOI will be deposited on acceptance.

Competing interests

The authors declare no competing interests.

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